

TASK-1

```
Switch>enab
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name sales
Switch(config-vlan)#exit
Switch(config)#valn 20
      ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 20
Switch(config-vlan)#name Support
Switch(config-vlan)#exit
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	sales	active	
20	Support	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

TASK-2

VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	sales	active	Fa0/1, Fa0/2, Fa0/3
20	Support	active	Fa0/4, Fa0/5
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Task-4

Scenario: Sales VLAN Cannot Ping Support VLAN

Network Setup

VLAN	Department	Device	IP Address
VLAN 10	Sales	PC-Sales	192.168.10.10/24
VLAN 20	Support	PC-Support	192.168.20.10/24

Ping Test

On the Sales PC:

```
C:\> ping 192.168.20.10
```

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Why the Ping Fails

The ping fails because **VLAN 10 (Sales)** and **VLAN 20 (Support)** are **different broadcast domains**. A Layer 2 switch forwards traffic only within the same VLAN and **cannot route packets between different VLANs**. Since there is **no Layer 3 device** (router or Layer 3 switch) configured to route traffic between the VLANs, the packets are dropped.

How to Allow Inter-VLAN Communication

To enable communication between the Sales and Support VLANs, configure **inter-VLAN routing** using either:

1. Router-on-a-Stick

- Connect the switch to a router using a **trunk link**.
- Create one **subinterface** on the router for each VLAN (e.g., VLAN 10 and VLAN 20).
- Assign each subinterface an IP address to act as the **default gateway**.

Example:

- VLAN 10 Gateway: **192.168.10.1**
- VLAN 20 Gateway: **192.168.20.1**

OR

2. Layer 3 Switch

- Create **SVIs (Switched Virtual Interfaces)** for each VLAN.
- Enable **IP routing** on the Layer 3 switch.

Result After Configuration

```
C:\> ping 192.168.20.10
```

```
Reply from 192.168.20.10: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.20.10: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.20.10: bytes=32 time=3ms TTL=128
```

```
Reply from 192.168.20.10: bytes=32 time=2ms TTL=128
```

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)